Evaluation report

Carefull transaction alert pipeline · model `gemini-3.1-flash-lite` · 291 transactions, 5 users, Jul–Oct 2024 · generated 2026-10-21

Bottom line

The system alerts exactly the three users who are being victimized (Margaret Thompson, Dorothy Williams, Helen Park) and none of the two users with legitimately high or unusual spending. Cheap rules find every case (recall 100%) but also flag 38 benign transactions; the LLM, reading each user's profile and history, removes them.

5 / 5
users classified correctly (alert / no alert)

32 / 32
suspicious transactions caught (100% recall)

0
false alarms after LLM validation (38 of 38 rule false positives removed)

$11
estimated LLM cost per month at 50,000 transactions

291 transactions → **70** flagged by rules (24%) → **32** confirmed by the LLM → **3** users alerted

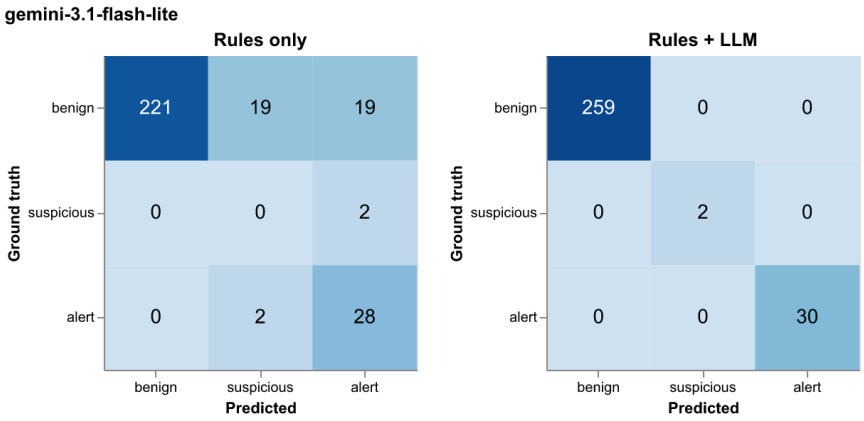
Read these numbers with caution

There is no held-out data. Rule thresholds were chosen against these same labels and the prompts were written after exploring these same five users. 32 positives come from only 3 incidents, so one error moves recall by ~3 points. The result shows the design works on these patterns, not how it generalizes.

Classification metrics

Flagged = suspicious or alert (anything a human should look at). **Alert** = only the class that notifies a trusted contact.

STAGE	FLAGGED P	FLAGGED R	FLAGGED F1	ALERT P	ALERT R	ALERT F1	ACCURACY
Rules only	0.46	1.00	0.63	0.57	0.93	0.71	0.86
Rules + LLM	1.00	1.00	1.00	1.00	1.00	1.00	1.00



Confusion matrices: rows = ground truth, columns = prediction.

Error analysis

Where the rule errors come from

RULE	FLAGGED	TRUE POS.	FALSE POS.	PRECISION	FP CLEARED BY LLM
Money mule	12	12	0	100%	0
Amount spike	51	21	30	41%	30
Weekly overspend	37	21	16	57%	16

USER	RULE FPS	MAIN CATEGORIES
Robert Chen	13	travel (7), transfer (4), shopping (1)
Frank Martinez	11	financial_services (5), transfer (4), groceries (1)
Margaret Thompson	5	utilities (2), groceries (2), dining (1)
Helen Park	5	housing (3), shopping (2)
Dorothy Williams	4	utilities (3), personal_care (1)

The errors are systematic, not random

Most rule false positives belong to **Robert Chen** and **Frank Martinez**: active users whose spending is high and variable by design (travel, golf, monthly transfers to family). Amount-based rules cannot tell "unusual for a statistic" from "unusual for this person"; the profile can. The money-mule rule is the only one that is precise on its own.

How the LLM removed them

- **Robert Chen** – BRITISH AIRWAYS PNR XKJD92 ($2,340, Amount spike, Weekly overspend): *"The transaction is a large travel expense, but it is consistent with the user's stated lifestyle of frequent travel and is supported by recent evidence of international activity (an Uber trip in London)."*
- **Helen Park** – IKEA PORTLAND OR ($580, Amount spike): *"The transaction at IKEA is consistent with the user's recent move to a new condo and her established shopping history in Portland. The amount is reasonable for home furnishings and does not indicate fraud."*
- **Frank Martinez** – ZELLE TO ISABELLA MARTINEZ ($500, Amount spike): *"While the $500 transfer to Isabella Martinez is larger than usual, it is consistent with Frank's established behavior of being generous with his family. Given his stable pension income, this is likely a planned gift rather than fraudulent activity."*

Severity: the LLM escalates as evidence accumulates

Transfers to a new payee. The rules get the severity wrong on 4 of 8; after validation, 0 remain. The LLM treats the first payment as something to monitor and escalates to an alert once the amounts keep growing.

DATE	AMOUNT	DESCRIPTION	RULES	LLM RISK (CONF.)	FINAL	TRUTH
Sep 05	$200	ZELLE PAYMENT TO KEVIN R	alert	medium (0.70)	suspicious	suspicious
Sep 12	$350	ZELLE TO KEVIN RICHARDS	alert	medium (0.80)	suspicious	suspicious
Sep 18	$500	ZELLE PAYMENT TO KEVIN RICH	alert	high (0.90)	alert	alert
Sep 25	$750	ZELLE TRANSFER KEVIN RICHARDS	alert	high (0.95)	alert	alert
Oct 02	$1,000	ZELLE PMT TO KEVIN R	alert	high (0.90)	alert	alert
Oct 08	$1,500	ZELLE PAYMENT KEVIN RICHARDS	alert	high (0.95)	alert	alert
Oct 15	$2,000	ZELLE TO KEVIN RICHARDS	suspicious	high (0.95)	alert	alert
Oct 22	$1,200	WIRE TRANSFER TO KEVIN RICHARDS	suspicious	high (0.95)	alert	alert

Remaining errors after validation

No remaining errors

No false negatives and no false positives after validation, and the alert / suspicious severity matches the ground truth on every flagged transaction.

Confidence and uncertainty handling

- LLM confidence ranged from 0.70 to 1.00 (median 0.95); 0 transactions went to human review.
- Because every validated decision was correct, this data **cannot tell whether confidence is calibrated**. The low-confidence path (< threshold → suspicious + human review) and the LLM-failure fallback (keep the rule label + review) were not exercised by real cases here.

What could fail on new data

- **Rule recall is the ceiling.** The LLM only sees what rules flag. A slow exploitation with flat amounts (e.g. $300 every week) would not trip the spike rule.
- **Weekly rule latency.** It uses the calendar week's total, so it fires up to 7 days late; a rolling 7-day window is needed in production.
- **Alias resolution is permissive.** The "same initial" rule would merge Marco and Maria Martinez.
- **Biographies help a lot.** "Wife Patricia manages household expenses" explains Robert's transfers. Users with thin profiles will produce more false alarms.

Model comparison

MODEL	STATUS	FLAGGED F1	ALERT F1	LATENCY P50	LATENCY P95	RUN COST	MONTHLY (50K TX)
gemini-3.1-flash-lite	measured	1.00	1.00	2.6 s	17.7 s	$0.064	$11
gemini-3.8-flash	estimated*	not run	not run	-	-	$0.179	$31

* Not run: on the free tier `gemini-3.8-flash` allows 20 requests/day and one run needs 94. The estimate applies its price to the tokens measured with `gemini-3.1-flash-lite`; it would be higher if the model spends thinking tokens. p95 latency is inflated by free-tier rate-limit retries; the median is representative of a single call.

> **Model choice**
> Validation is a reading-comprehension task over ~2k tokens of structured context, and the lite model already makes no errors here, so a bigger model can only add cost and latency on this data. A larger model is worth testing on harder, ambiguous cases (thin profiles, partial patterns), where the comparison would actually discriminate.

Cost projection

$0.0009
per validation call

$0.00022
per transaction (all steps)

$11.07
per month at 50,000 tx (real time)

$5.54
per month with Batch API

LLM STEP	CALLS	AVG TOKENS IN / OUT	COST PER CALL	WHEN
Counterparty extraction	24	426 / 28	$0.00015	once per unique description (cached)
Risk validation	70	1,953 / 254	$0.00087	once per rule-flagged transaction

Cost scales with how many transactions the rules flag, not with total volume: only 24% of transactions reach the validation step. Monthly cost at other flag rates: 5% → **$2.79**, 10% → **$4.96**. Prices: Gemini API standard tier, October 2026.

Where to optimize

- **Lower the flag rate first.** 38 of 70 validations were benign. Tightening the rules (as with the weekly rule: 159 → 70 flags with no recall loss) saves more than any model change.
- **One call per incident, not per transaction.** Helen's 12 foreign purchases are one story; validating the pattern once would cut those calls by ~10x.
- **Batch API** for the non-urgent weekly checks (50% cheaper) and **prompt caching** for the fixed system prompt and the user's profile, which repeat on every call of a user.
- At these prices the LLM is not the main cost driver; analyst time on false alarms is. Keep the model that minimizes false alarms, not the cheapest one.